Hand Gesture based Controlled Car

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Introduction

- Hand gesture-based control is an innovative way to interact with technology, and cars are no exception. This technology uses cameras and sensors to detect and interpret these gestures, the system can translate them into commands that control the car's various functions.
- This technology offers a more intuitive and potentially safer way to drive, as it reduces the need for drivers to physically manipulate the steering wheel, gear shift, and other controls. Additionally, it can be particularly beneficial for drivers with disabilities who may find traditional controls challenging.

Overview of Project

•**Gesture Recognition**: The device uses a gesture-based interface, allowing users to control its movement (forward, backward, left, and right) using predefined hand signals.

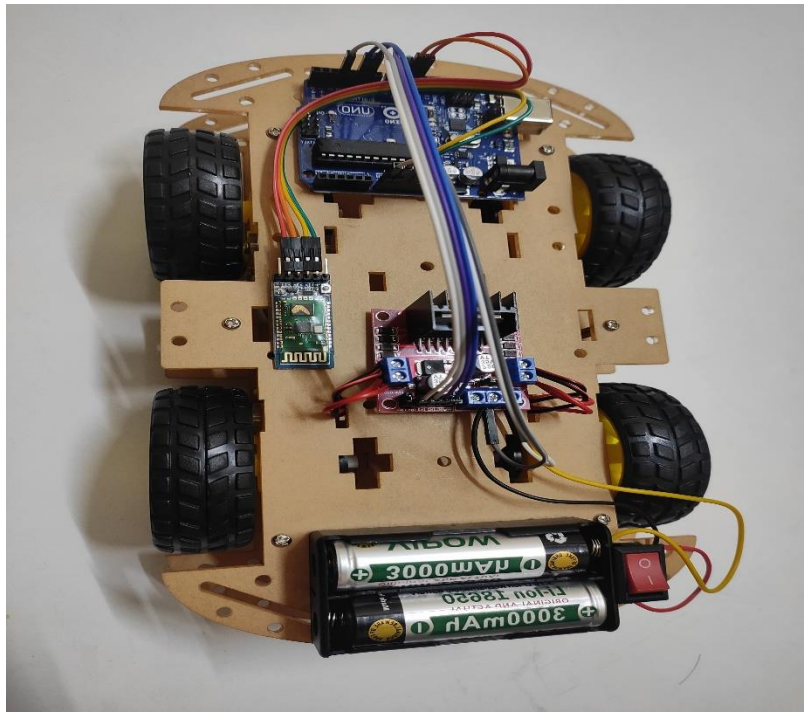
•**Wireless Communication**: The device is equipped with a wireless module (such as Bluetooth or Wi-Fi) to receive commands from a hand gesture recognition system, ensuring smooth real-time control.

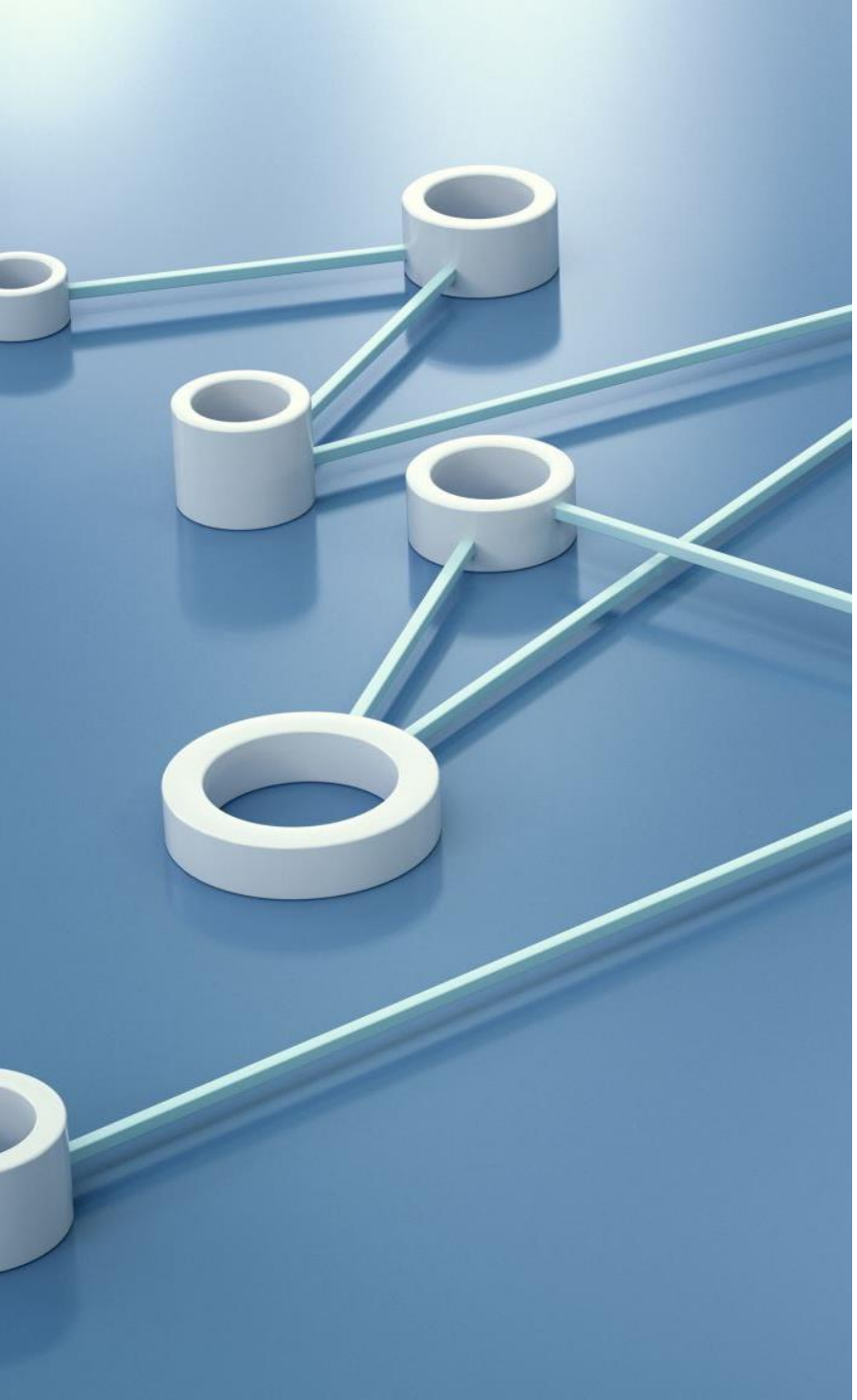
•**Responsive Control System**: The control system is responsive to quick hand gestures, making it highly interactive and user-friendly.

•**Microcontroller Integration**: At its core, a microcontroller (like Arduino or Raspberry Pi) processes the signals and controls the motors and other mechanical components.

Car

- We have assembled the car chassis kit, fixed motors, wires, Arduino UNO, L298N Motor drive and HC-05 Bluetooth module.

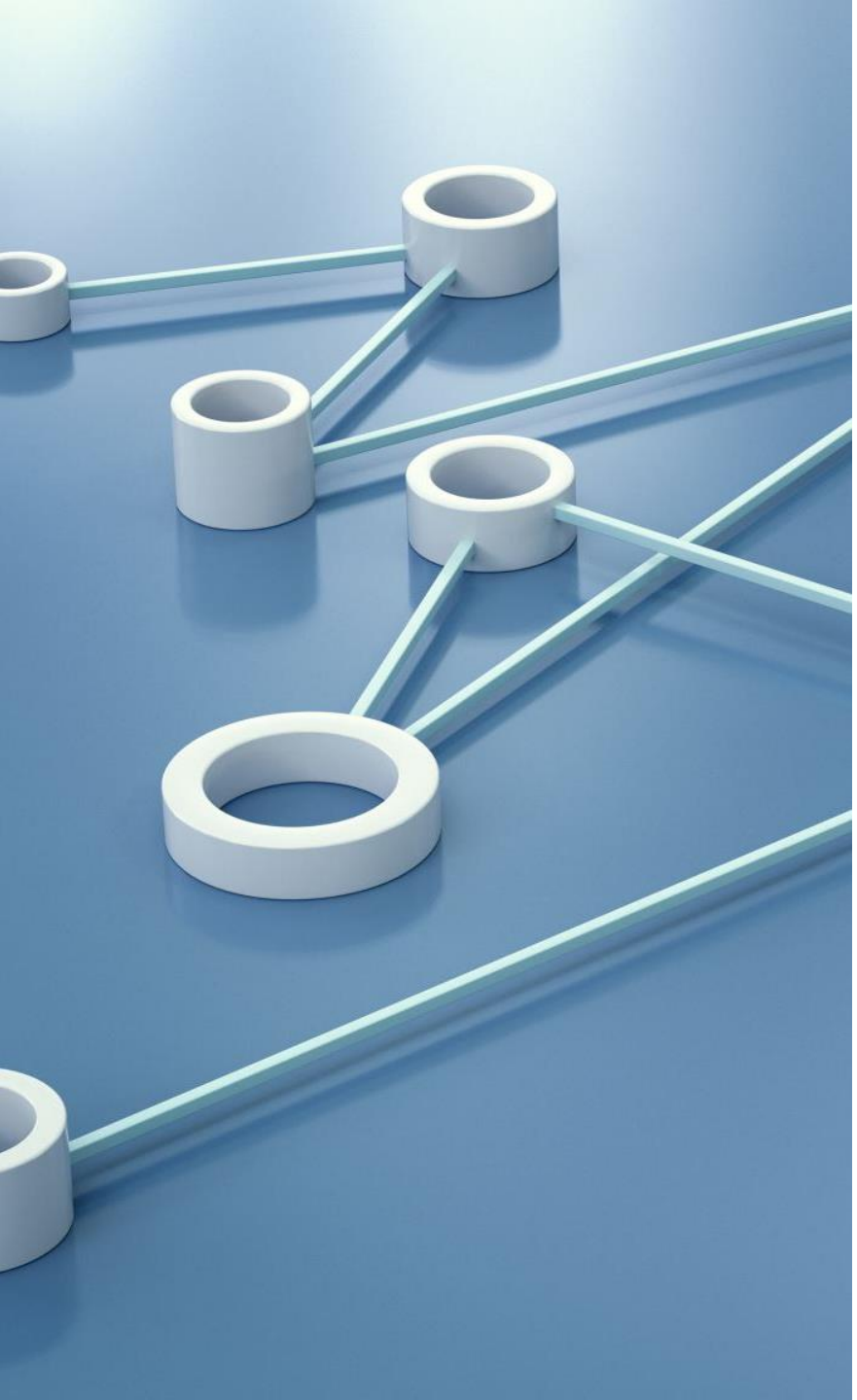




Connections

From Motors to Motor Drive

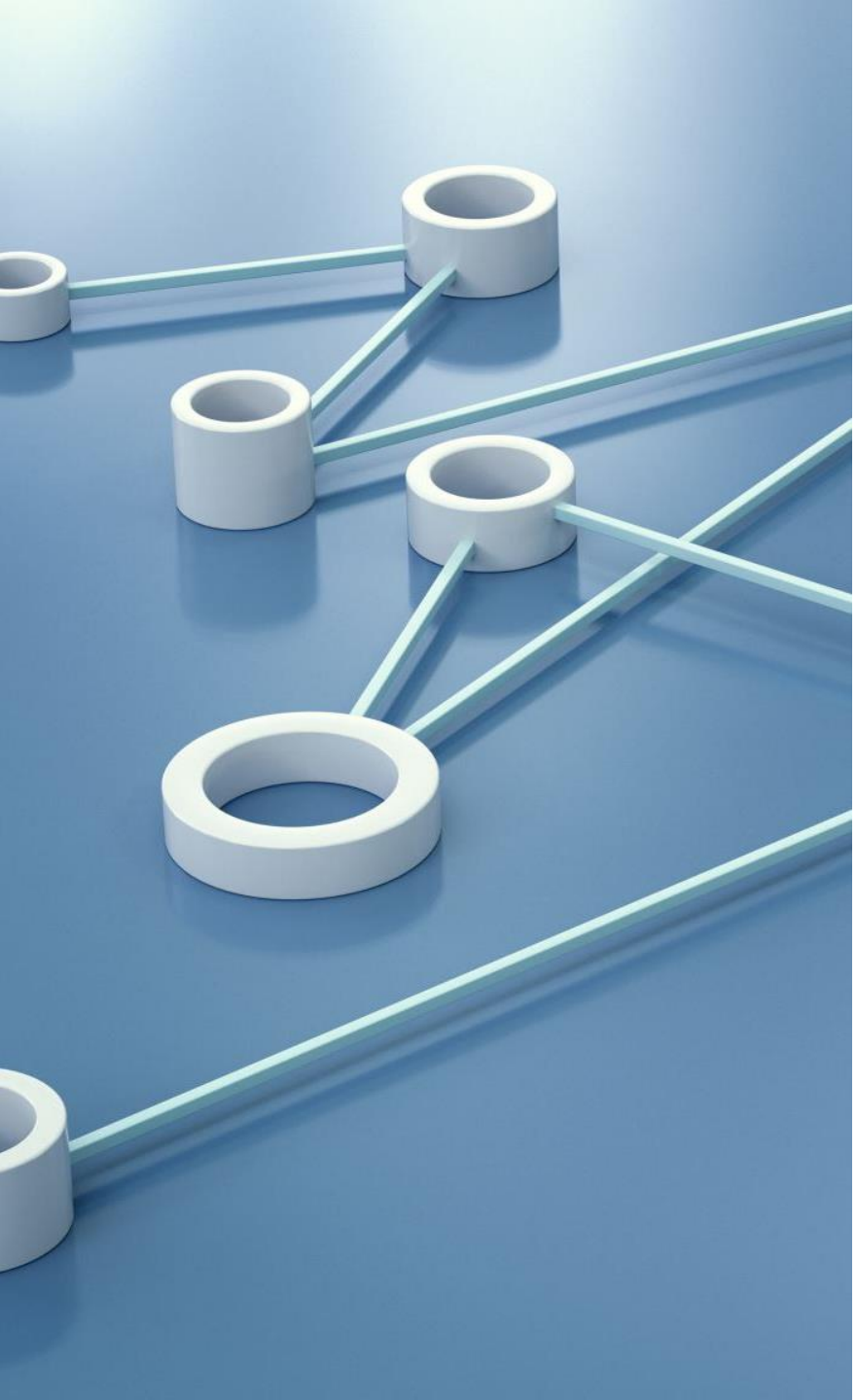
Motor Wire	L298N Motor Drive
The two red wires on the two left motors.	OUT2
The two black wires on the two left motors.	OUT1
The two red wires on the two Right motors.	OUT3
The two black wires on the two Right motors.	OUT4



Connections

From Motor Drive to Arduino UNO

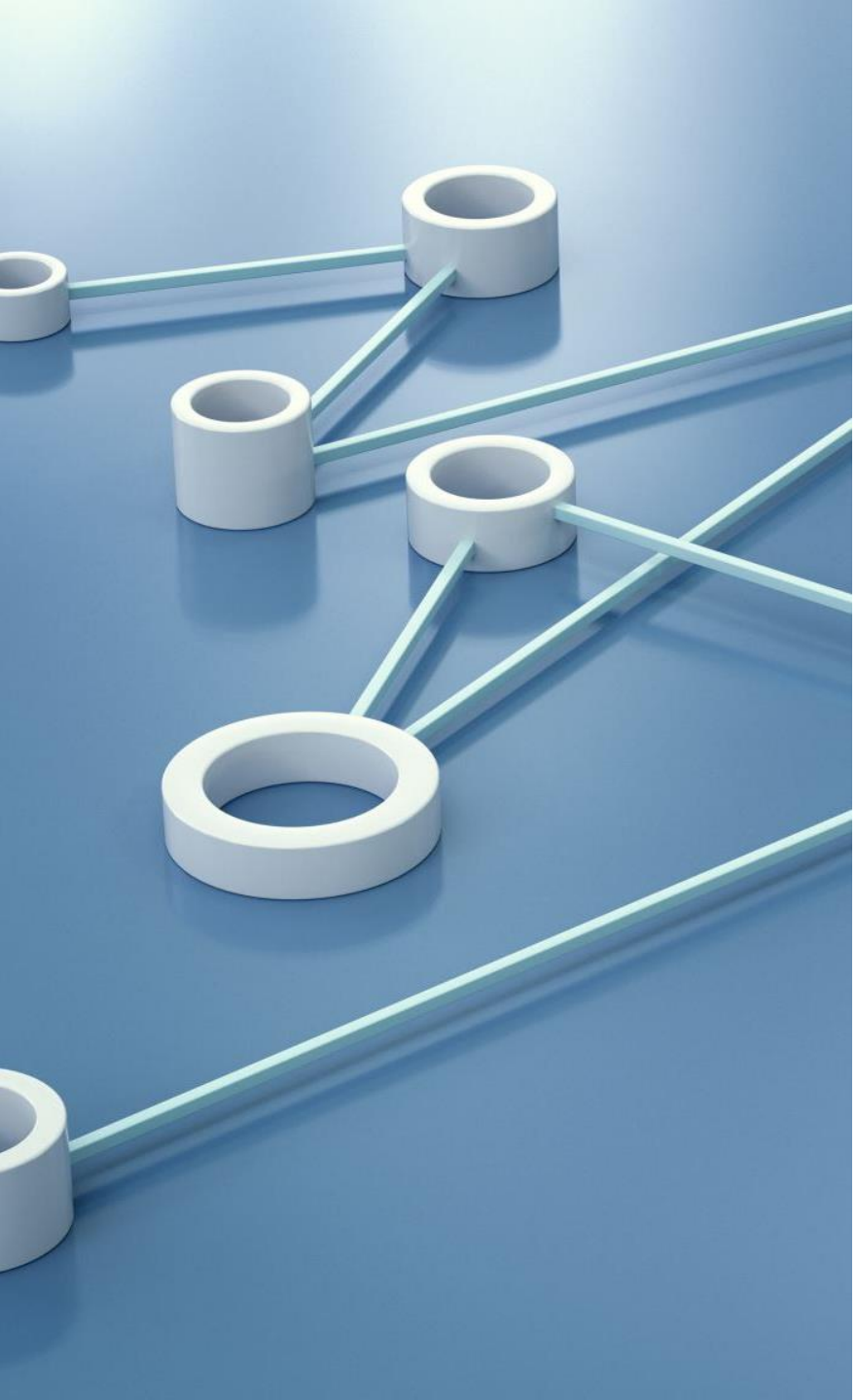
L298N Motor Drive	Arduino UNO
IN1	5
IN2	6
IN3	3
IN4	9



Connections

From Arduino UNO to HC-05 Bluetooth (Slave) Module

Slave Bluetooth	Arduino UNO
RXD	11
TXD	10
GND	GND
VCC	5V



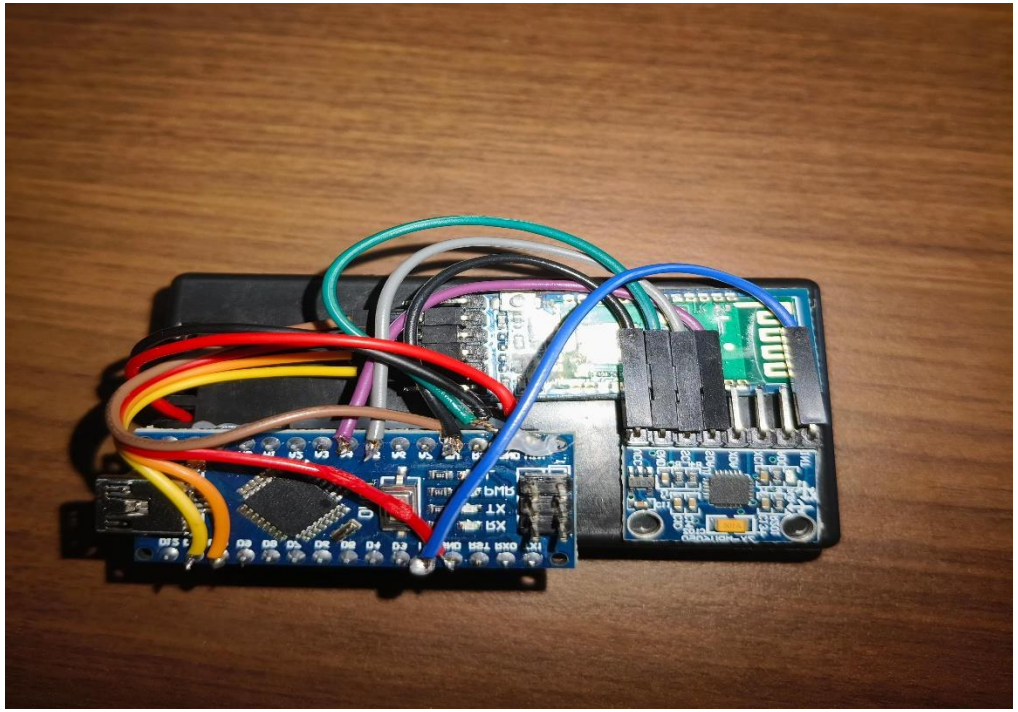
Connections

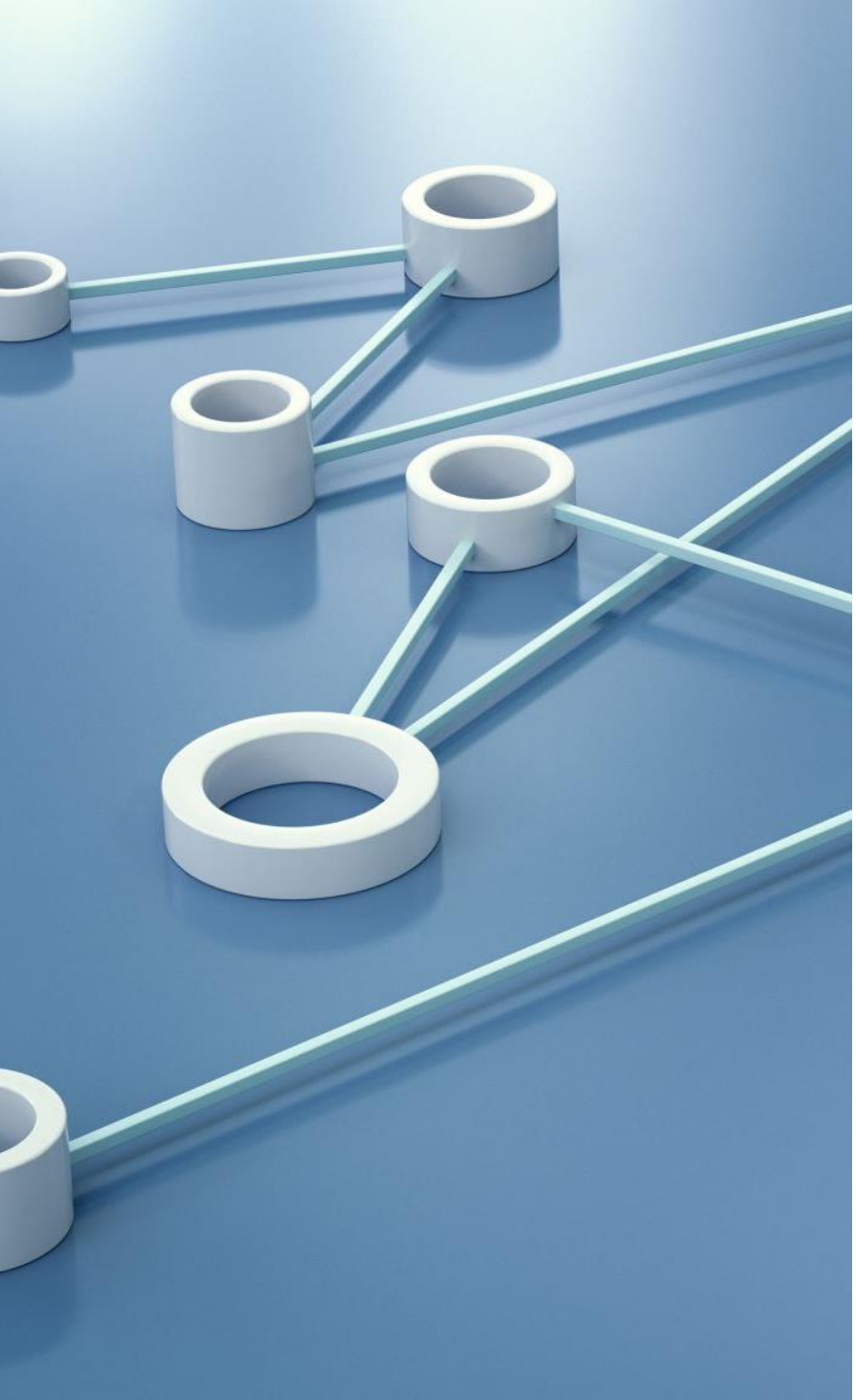
Power Supply connections

Arduino UNO	L298N Motor Drive	Power Supply
VIN	12V Connector	Red wire (+)
GND	GND Connector	Black wire (-)

Hand Controller

- We have connected HC-05, MPU6050(For Gesture recognition) and Arduino Nano with power supply.

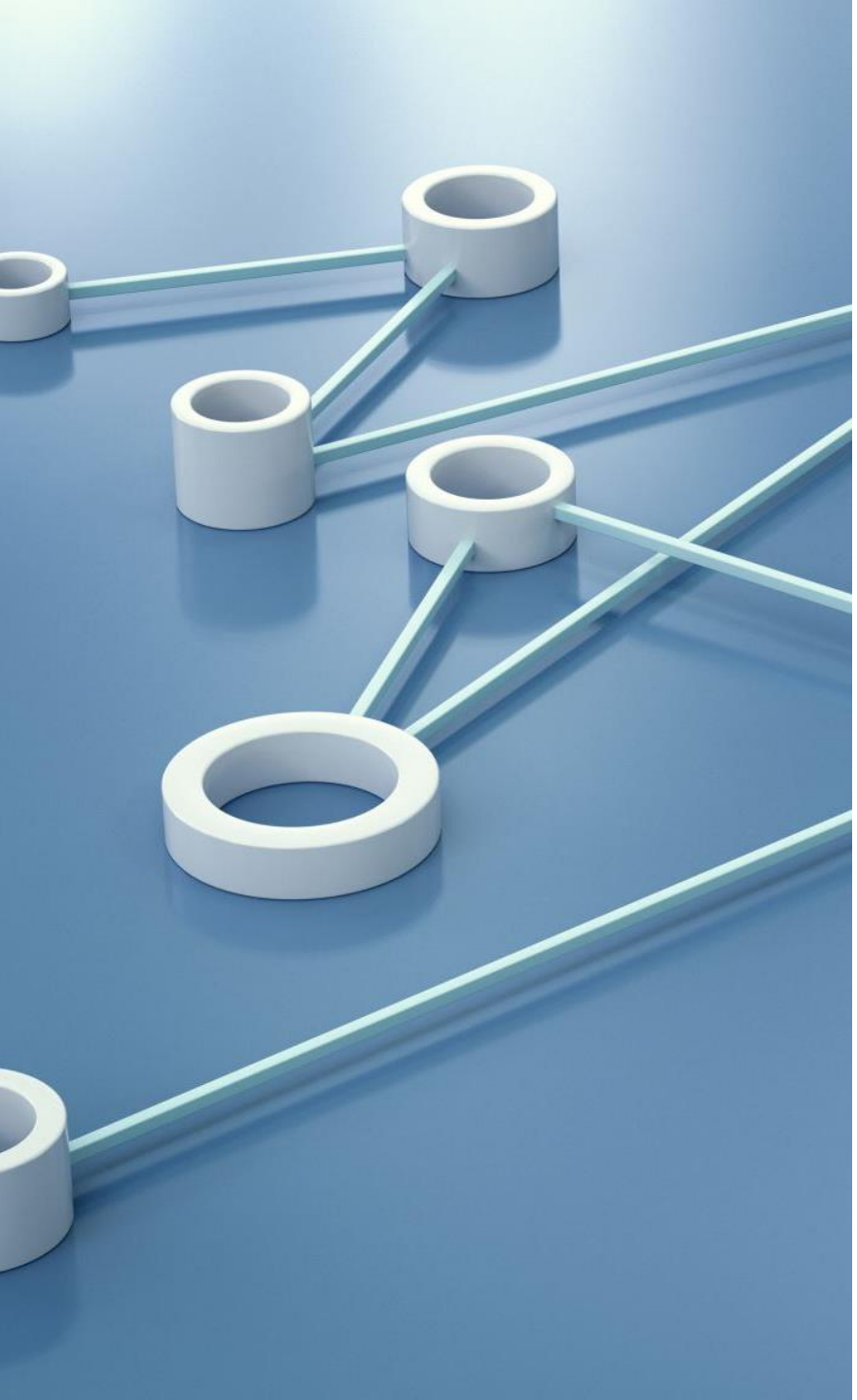




Connections

From MPU 6050

MPU 6050	Arduino NANO
SDA	A4
SCL	A5
INT	2
GND	GND
VCC	5V



Connections

From HC-05 Bluetooth (Master) Module

Master Bluetooth Module	ARDUINO NANO
RXD	11
TXD	10
GND	GND
VCC	5v

Arduino Codes

https://drive.google.com/drive/folders/1GwVSqrxESujlEakbiv7-iwvA5tuFQ_dL?usp=sharing

Key Points from Arduino configuration(Transmitter)

- Utilizes an MPU6050 sensor to detect the orientation and movement of the user's hand.
- The code calculates the yaw, pitch, and roll angles from the sensor data to determine the direction of movement.
- Implements a SoftwareSerial connection to communicate with an HC-05 Bluetooth module.
- Sends control commands ('F', 'B', 'L', 'R', 'S') to the receiving Arduino based on gesture input.
- Uses an interrupt to signal when new data is available from the MPU6050 sensor, ensuring timely processing of gesture inputs.
- Utilizes a FIFO buffer to manage incoming data from the MPU6050 and check overflow conditions.
- Converts quaternion data from the MPU6050 to yaw, pitch, and roll values.
- Maps these values to a specific range for control decisions.
- Implements a simple LED blinking mechanism to indicate system status and data processing.
- Serial communication is used for debugging and monitoring values of pitch, roll, and gesture control outputs in real-time.
- The code is organized into functions (e.g., forward(), reverse(), stopCar(), etc.), making it easier to read.

Key Points from Arduino configuration(Receiver)

- Controls the car's movement through an L298N motor driver with defined input pins for forward, backward, left, right, and stop functions.
- Uses digitalWrite and analogWrite functions to control motor speed and direction.
- Receives control commands ('F', 'B', 'L', 'R', 'S') from the transmitter via the HC-05 Bluetooth module.
- Implements functions to control motor actions (e.g., forward, reverse, left, right, stop).
- Organizes code into functions for better readability and maintenance.

Advantages

- **Intuitive Interaction**: Gestures are natural and easy to learn, making it simple for users to interact with devices without needing complex input methods.
- **Hands-Free Operation**: They allow users to control devices without physically touching them, which is convenient in situations where hands are occupied or cleanliness is essential.
- **Accessibility**: These devices can enhance accessibility for people with disabilities who may struggle with traditional input methods like keyboards or touchscreens.
- **Increased Efficiency**: Quick, simple gestures can often perform tasks faster than traditional input methods, streamlining user interaction.
- **Enhanced User Experience**: Gesture controls can create a more immersive, modern, and enjoyable user experience, especially in virtual reality, gaming, or smart home environments.
- **Contactless Control:** Ideal for situations where hygiene is important, such as medical environments, public kiosks, or industrial settings, minimizing the need for direct contact.

Thank You